

# Placement DSA Problem Sheet

Curated, ordered problem list for campus placement preparation — ~373 problems across 14 phases, sourced primarily from LeetCode. Solve in phase order; mark difficulty and revisit after 1 day / 7 days / 30 days.

## Phase 0: Programming Foundation

| No. | Problem                             | Difficulty | Platform / Problem # |
|-----|-------------------------------------|------------|----------------------|
| 1   | Two Sum                             | Easy       | LeetCode #1          |
| 2   | Remove Duplicates from Sorted Array | Easy       | LeetCode #26         |
| 3   | Best Time to Buy and Sell Stock     | Easy       | LeetCode #121        |
| 4   | Contains Duplicate                  | Easy       | LeetCode #217        |
| 5   | Majority Element                    | Easy       | LeetCode #169        |
| 6   | Rotate Array                        | Medium     | LeetCode #189        |
| 7   | Move Zeroes                         | Easy       | LeetCode #283        |
| 8   | Merge Sorted Array                  | Easy       | LeetCode #88         |
| 9   | Valid Anagram                       | Easy       | LeetCode #242        |
| 10  | Valid Palindrome                    | Easy       | LeetCode #125        |
| 11  | Reverse String                      | Easy       | LeetCode #344        |
| 12  | Reverse Words in a String           | Medium     | LeetCode #151        |
| 13  | Longest Common Prefix               | Easy       | LeetCode #14         |
| 14  | Roman to Integer                    | Easy       | LeetCode #13         |
| 15  | Integer to Roman                    | Medium     | LeetCode #12         |

## Phase 1: Two Pointers & Sliding Window

| No. | Problem                                        | Difficulty | Platform / Problem # |
|-----|------------------------------------------------|------------|----------------------|
| 16  | Two Sum II - Input Array Is Sorted             | Medium     | LeetCode #167        |
| 17  | Container With Most Water                      | Medium     | LeetCode #11         |
| 18  | 3Sum                                           | Medium     | LeetCode #15         |
| 19  | 4Sum                                           | Medium     | LeetCode #18         |
| 20  | Sort Colors                                    | Medium     | LeetCode #75         |
| 21  | Remove Nth Node From End of List               | Medium     | LeetCode #19         |
| 22  | Squares of a Sorted Array                      | Easy       | LeetCode #977        |
| 23  | Backspace String Compare                       | Easy       | LeetCode #844        |
| 24  | Maximum Average Subarray I                     | Easy       | LeetCode #643        |
| 25  | Longest Substring Without Repeating Characters | Medium     | LeetCode #3          |
| 26  | Minimum Size Subarray Sum                      | Medium     | LeetCode #209        |

| No. | Problem                                 | Difficulty | Platform / Problem # |
|-----|-----------------------------------------|------------|----------------------|
| 27  | Permutation in String                   | Medium     | LeetCode #567        |
| 28  | Find All Anagrams in a String           | Medium     | LeetCode #438        |
| 29  | Minimum Window Substring                | Hard       | LeetCode #76         |
| 30  | Longest Repeating Character Replacement | Medium     | LeetCode #424        |
| 31  | Fruit Into Baskets                      | Medium     | LeetCode #904        |
| 32  | Max Consecutive Ones III                | Medium     | LeetCode #1004       |

## Phase 2: HashMap + Prefix Sum

| No. | Problem                      | Difficulty | Platform / Problem # |
|-----|------------------------------|------------|----------------------|
| 33  | Group Anagrams               | Medium     | LeetCode #49         |
| 34  | Top K Frequent Elements      | Medium     | LeetCode #347        |
| 35  | Happy Number                 | Easy       | LeetCode #202        |
| 36  | Isomorphic Strings           | Easy       | LeetCode #205        |
| 37  | Word Pattern                 | Easy       | LeetCode #290        |
| 38  | Longest Consecutive Sequence | Medium     | LeetCode #128        |
| 39  | Subarray Sum Equals K        | Medium     | LeetCode #560        |
| 40  | Continuous Subarray Sum      | Medium     | LeetCode #523        |
| 41  | Range Sum Query - Immutable  | Easy       | LeetCode #303        |
| 42  | Product of Array Except Self | Medium     | LeetCode #238        |
| 43  | Find Pivot Index             | Easy       | LeetCode #724        |

## Phase 3: Binary Search

| No. | Problem                                 | Difficulty | Platform / Problem # |
|-----|-----------------------------------------|------------|----------------------|
| 44  | Binary Search                           | Easy       | LeetCode #704        |
| 45  | Search Insert Position                  | Easy       | LeetCode #35         |
| 46  | First Bad Version                       | Easy       | LeetCode #278        |
| 47  | Search in Rotated Sorted Array          | Medium     | LeetCode #33         |
| 48  | Find Peak Element                       | Medium     | LeetCode #162        |
| 49  | Koko Eating Bananas                     | Medium     | LeetCode #875        |
| 50  | Capacity To Ship Packages Within D Days | Medium     | LeetCode #1011       |
| 51  | Split Array Largest Sum                 | Hard       | LeetCode #410        |
| 52  | Median of Two Sorted Arrays             | Hard       | LeetCode #4          |
| 53  | Find Minimum in Rotated Sorted Array    | Medium     | LeetCode #153        |

## Phase 4: Linked List

| No. | Problem                            | Difficulty | Platform / Problem # |
|-----|------------------------------------|------------|----------------------|
| 54  | Reverse Linked List                | Easy       | LeetCode #206        |
| 55  | Middle of the Linked List          | Easy       | LeetCode #876        |
| 56  | Linked List Cycle                  | Easy       | LeetCode #141        |
| 57  | Linked List Cycle II               | Medium     | LeetCode #142        |
| 58  | Merge Two Sorted Lists             | Easy       | LeetCode #21         |
| 59  | Merge k Sorted Lists               | Hard       | LeetCode #23         |
| 60  | Remove Duplicates from Sorted List | Easy       | LeetCode #83         |
| 61  | Intersection of Two Linked Lists   | Easy       | LeetCode #160        |
| 62  | Reverse Linked List II             | Medium     | LeetCode #92         |
| 63  | LRU Cache                          | Medium     | LeetCode #146        |

## Phase 5: Stack & Queue

| No. | Problem                          | Difficulty | Platform / Problem # |
|-----|----------------------------------|------------|----------------------|
| 64  | Valid Parentheses                | Easy       | LeetCode #20         |
| 65  | Min Stack                        | Medium     | LeetCode #155        |
| 66  | Next Greater Element I           | Easy       | LeetCode #496        |
| 67  | Daily Temperatures               | Medium     | LeetCode #739        |
| 68  | Asteroid Collision               | Medium     | LeetCode #735        |
| 69  | Decode String                    | Medium     | LeetCode #394        |
| 70  | Evaluate Reverse Polish Notation | Medium     | LeetCode #150        |
| 71  | Largest Rectangle in Histogram   | Hard       | LeetCode #84         |
| 72  | Trapping Rain Water              | Hard       | LeetCode #42         |
| 73  | Online Stock Span                | Medium     | LeetCode #901        |
| 74  | Sum of Subarray Minimums         | Medium     | LeetCode #907        |
| 75  | Implement Queue using Stacks     | Easy       | LeetCode #232        |
| 76  | Implement Stack using Queues     | Easy       | LeetCode #225        |
| 77  | Design Circular Queue            | Medium     | LeetCode #622        |

## Phase 6: Trees

| No. | Problem                        | Difficulty | Platform / Problem # |
|-----|--------------------------------|------------|----------------------|
| 78  | Binary Tree Preorder Traversal | Easy       | LeetCode #144        |

| No. | Problem                                  | Difficulty | Platform / Problem # |
|-----|------------------------------------------|------------|----------------------|
| 79  | Binary Tree Inorder Traversal            | Easy       | LeetCode #94         |
| 80  | Binary Tree Postorder Traversal          | Easy       | LeetCode #145        |
| 81  | Binary Tree Level Order Traversal        | Medium     | LeetCode #102        |
| 82  | Binary Tree Zigzag Level Order Traversal | Medium     | LeetCode #103        |
| 83  | Validate Binary Search Tree              | Medium     | LeetCode #98         |
| 84  | Lowest Common Ancestor of a BST          | Medium     | LeetCode #235        |
| 85  | Kth Smallest Element in a BST            | Medium     | LeetCode #230        |
| 86  | Insert into a Binary Search Tree         | Medium     | LeetCode #701        |
| 87  | Delete Node in a BST                     | Medium     | LeetCode #450        |
| 88  | Diameter of Binary Tree                  | Easy       | LeetCode #543        |
| 89  | Balanced Binary Tree                     | Easy       | LeetCode #110        |
| 90  | Maximum Depth of Binary Tree             | Easy       | LeetCode #104        |
| 91  | Path Sum                                 | Easy       | LeetCode #112        |
| 92  | Same Tree                                | Easy       | LeetCode #100        |
| 93  | Symmetric Tree                           | Easy       | LeetCode #101        |
| 94  | Serialize and Deserialize Binary Tree    | Hard       | LeetCode #297        |
| 95  | Binary Tree Right Side View              | Medium     | LeetCode #199        |

## Phase 7: Heap

| No. | Problem                         | Difficulty | Platform / Problem #        |
|-----|---------------------------------|------------|-----------------------------|
| 96  | Kth Largest Element in an Array | Medium     | LeetCode #215               |
| 97  | Top K Frequent Elements         | Medium     | LeetCode #347               |
| 98  | Merge k Sorted Lists            | Hard       | LeetCode #23                |
| 99  | Last Stone Weight               | Easy       | LeetCode #1046              |
| 100 | Task Scheduler                  | Medium     | LeetCode #621               |
| 101 | Meeting Rooms II                | Medium     | LeetCode #253 (GFG/Premium) |
| 102 | Reorganize String               | Medium     | LeetCode #767               |

## Phase 8: Graphs

| No. | Problem           | Difficulty | Platform / Problem # |
|-----|-------------------|------------|----------------------|
| 103 | Number of Islands | Medium     | LeetCode #200        |
| 104 | Rotting Oranges   | Medium     | LeetCode #994        |
| 105 | Clone Graph       | Medium     | LeetCode #133        |

| No. | Problem                               | Difficulty | Platform / Problem # |
|-----|---------------------------------------|------------|----------------------|
| 106 | Word Ladder                           | Hard       | LeetCode #127        |
| 107 | Flood Fill                            | Easy       | LeetCode #733        |
| 108 | Surrounded Regions                    | Medium     | LeetCode #130        |
| 109 | Course Schedule                       | Medium     | LeetCode #207        |
| 110 | Pacific Atlantic Water Flow           | Medium     | LeetCode #417        |
| 111 | Number of Provinces                   | Medium     | LeetCode #547        |
| 112 | Network Delay Time                    | Medium     | LeetCode #743        |
| 113 | Course Schedule II (Topological Sort) | Medium     | LeetCode #210        |
| 114 | Redundant Connection (Union Find)     | Medium     | LeetCode #684        |

## Phase 9: Backtracking

| No. | Problem                               | Difficulty | Platform / Problem # |
|-----|---------------------------------------|------------|----------------------|
| 115 | Subsets                               | Medium     | LeetCode #78         |
| 116 | Subsets II                            | Medium     | LeetCode #90         |
| 117 | Permutations                          | Medium     | LeetCode #46         |
| 118 | Permutations II                       | Medium     | LeetCode #47         |
| 119 | Combination Sum                       | Medium     | LeetCode #39         |
| 120 | Combination Sum II                    | Medium     | LeetCode #40         |
| 121 | N-Queens                              | Hard       | LeetCode #51         |
| 122 | Generate Parentheses                  | Medium     | LeetCode #22         |
| 123 | Letter Combinations of a Phone Number | Medium     | LeetCode #17         |
| 124 | Palindrome Partitioning               | Medium     | LeetCode #131        |

## Phase 10: Greedy

| No. | Problem                   | Difficulty | Platform / Problem # |
|-----|---------------------------|------------|----------------------|
| 125 | Jump Game                 | Medium     | LeetCode #55         |
| 126 | Jump Game II              | Medium     | LeetCode #45         |
| 127 | Gas Station               | Medium     | LeetCode #134        |
| 128 | Candy                     | Hard       | LeetCode #135        |
| 129 | Non-overlapping Intervals | Medium     | LeetCode #435        |
| 130 | Merge Intervals           | Medium     | LeetCode #56         |
| 131 | Insert Interval           | Medium     | LeetCode #57         |
| 132 | Partition Labels          | Medium     | LeetCode #763        |

## Phase 11: Dynamic Programming

| No. | Problem                        | Difficulty | Platform / Problem # |
|-----|--------------------------------|------------|----------------------|
| 133 | Climbing Stairs                | Easy       | LeetCode #70         |
| 134 | House Robber                   | Medium     | LeetCode #198        |
| 135 | House Robber II                | Medium     | LeetCode #213        |
| 136 | Coin Change                    | Medium     | LeetCode #322        |
| 137 | Decode Ways                    | Medium     | LeetCode #91         |
| 138 | Unique Paths                   | Medium     | LeetCode #62         |
| 139 | Minimum Path Sum               | Medium     | LeetCode #64         |
| 140 | Edit Distance                  | Hard       | LeetCode #72         |
| 141 | Longest Common Subsequence     | Medium     | LeetCode #1143       |
| 142 | Longest Increasing Subsequence | Medium     | LeetCode #300        |
| 143 | Partition Equal Subset Sum     | Medium     | LeetCode #416        |
| 144 | Target Sum                     | Medium     | LeetCode #494        |
| 145 | Word Break                     | Medium     | LeetCode #139        |
| 146 | Maximum Product Subarray       | Medium     | LeetCode #152        |

## Phase 12: Bit Manipulation

| No. | Problem          | Difficulty | Platform / Problem # |
|-----|------------------|------------|----------------------|
| 147 | Single Number    | Easy       | LeetCode #136        |
| 148 | Missing Number   | Easy       | LeetCode #268        |
| 149 | Counting Bits    | Easy       | LeetCode #338        |
| 150 | Reverse Bits     | Easy       | LeetCode #190        |
| 151 | Number of 1 Bits | Easy       | LeetCode #191        |

## Phase 13: Tries

| No. | Problem                                    | Difficulty | Platform / Problem # |
|-----|--------------------------------------------|------------|----------------------|
| 152 | Implement Trie (Prefix Tree)               | Medium     | LeetCode #208        |
| 153 | Design Add and Search Words Data Structure | Medium     | LeetCode #211        |
| 154 | Word Search II                             | Hard       | LeetCode #212        |

## Phase 14: Revision & Mock Interviews

After completing the core topics above: solve 3-4 mixed problems daily, take one timed online assessment every week, revisit problems you couldn't solve independently, and practice explaining your approach aloud before coding.

## Recommended Platform Priority

| Priority | Platform          | Why                                                    |
|----------|-------------------|--------------------------------------------------------|
| ■■■■■    | LeetCode          | Best for OAs and product-company interviews            |
| ■■■■     | GeeksforGeeks     | Good explanations, useful for service-company practice |
| ■■■■     | NeetCode          | Excellent ordering with video explanations             |
| ■■■■     | Striver A2Z Sheet | Covers fundamentals systematically                     |
| ■■■      | CodeStudio        | Optional supplementary practice                        |
| ■■       | InterviewBit      | Optional                                               |
| ■        | HackerRank        | Mainly for language-basics practice                    |

## Month-wise Suggested Timeline

| Month            | Focus                                                              |
|------------------|--------------------------------------------------------------------|
| August           | Arrays, Strings, Hashing, Two Pointers, Sliding Window             |
| September        | Binary Search, Stack, Queue, Linked List, Trees                    |
| October          | Graphs, Heap, Greedy, DP basics, OA practice                       |
| Placement Season | Mixed mocks, company-tagged problems, contests, interview practice |